# Project Abstract

## Analog Electronics Course Project

## Project Title

Design of a 1.2V Two-Stage Miller-Compensated CMOS Operational Amplifier using $g_m/I_D$ Methodology in 90nm Technology

## Objectives of the Project

To design, characterize, and simulate a stable two-stage Miller-compensated operational amplifier at the transistor level using a 90nm CMOS process. Due to severe short-channel effects at this technology node, the project aims to abandon traditional square-law hand calculations in favor of the $g_m/I_D$ sizing methodology. The objective is to achieve a balanced trade-off between open-loop gain, bandwidth, and power consumption while ensuring strict phase margin requirements for stability.

## Design Specifications

- **Technology Node:** 90nm CMOS (GPDK090)
- **Power Supply ($V_{DD}$):** 1.2 V
- **Open-Loop DC Gain:** $\geq$ 50 dB
- **Unity Gain Bandwidth (UGB):** $\geq$ 20 MHz
- **Phase Margin:** $\geq 60^\circ$
- **Slew Rate:** $\geq$ 10 V/$\mu$s
- **Load Capacitance ($C_L$):** 1 pF
- **Power Dissipation:** $\leq$ 1 mW

## Design Methodology/Approach

1. **Device Characterization:** Generate look-up tables and plots (e.g., $g_m/I_D$ vs. $I_D/W$, $g_m/I_D$ vs. $g_m r_o$) for single NMOS and PMOS devices in Cadence Virtuoso to map actual physical behavior.
2. **Architecture Selection:** Utilize a standard two-stage topology with a differential input pair, active current mirror load, and a common-source second stage.
3. **Transistor Sizing:** Calculate the required tail current based on the target Slew Rate and load capacitance. Use the extracted $g_m/I_D$ charts to determine the exact channel widths ($W$) required for the gain and bandwidth targets while operating in moderate-to-strong inversion.

4. **Compensation Design:** Size the Miller capacitor ($C_c$) to split the poles and introduce a nulling resistor ($R_z$) if necessary to cancel the right-half-plane zero, ensuring a phase margin of at least 60°.

5. **Simulation & Verification:** Perform DC, AC, and Transient analyses using Cadence ADE to verify all specifications are met, iterating on transistor geometries as needed.

## EDA Tools/Software to be Used

- Cadence Virtuoso (Schematic Editor, ADE L/XL for simulation)